

Return to play

with IAN SHRIER

*Perhaps the most common question for every active but injured patient is
'When can I return to play?'*

Return-to-play (RTP) decisions can be difficult and complex.¹ The clinician is expected to provide an opinion based on a large number of factors including the history of the injury, physical examination, type of injury, rehabilitation, type of activity, psychological state, competitive level and ability to protect the injury. Further, these decisions do not occur in a vacuum. The athlete may receive advice from family, friends, coaches, other clinicians and, in the case of elite athletes, agents.

Because determining prognosis is difficult and frequently subjective, some disagreements regarding RTP will always occur.² Differing sociocultural and clinical perspectives of the medical doctor, coach, athlete and others lead to a high potential for conflict. Apart from being challenging and an unpleasant experience, such conflicts are believed to lead to a number of negative scenarios including:

1. miscommunication
2. loss of trust
3. potential litigation
4. declines in sport participation rates—some individuals never 'get back in the game' due to fear of re-injury (despite acceptable levels of risk)
5. serious medical complications—some athletes return to activity while still at unacceptable levels of risk for subsequent sport-related injury.³⁻⁶

Two important factors may minimise development of such conflict: a formal structure or process outlining how an actual RTP decision should be made and a formal process to guide the interactions between individuals who contribute in any way to the RTP process. It is important to note that in some cultures the clinician is

expected to determine the reasonable course of action for a seemingly independent third party—the athlete—who is otherwise capable of making autonomous decisions. In other cultures shared decision making may be promoted by the medical community in general, but the clinician is usually still considered legally responsible for any consequences of the decision. Regardless of the cultural context in which one is situated, having a transparent framework for arriving at RTP decisions should help minimise any conflict that might arise.

The purpose of this chapter is to provide such a framework. In 2010, we published a 3-Step framework⁷ developed specifically for RTP decision making. The framework is consistent with clinicians' beliefs independent of country of practice or clinician specialty,⁸ and represents a framework to help organise complex information; it should not be interpreted as proscriptive. After feedback from clinicians, this has now evolved into a revised framework called the Strategic Assessment of Risk and Risk Tolerance (StARRT).⁹

The StARRT framework is based on causal relationships and considers that differences in RTP decisions are partly due to differences in (1) risk assessment and (2) risk tolerance. Differences in risk assessment could be decreased through the application of research and knowledge dissemination leading to improved evidence-based practice. Differences in risk tolerance are based on societal and personal values, which are independent of research and evidence-based practice. This may be part of the reason why different participants presented with the same information come to different RTP decisions.¹⁰ Finally, the framework's general approach means that it can be applied by whoever represents the decision-making

authority: a clinician, non-clinician or a multidisciplinary team.

STRATEGIC ASSESSMENT OF RISK AND RISK TOLERANCE FRAMEWORK FOR RTP DECISION MAKING

The StARRT framework for RTP decisions is illustrated in Figure 19.1. Rational decision making requires weighing the benefits and risks of different alternatives.¹¹⁻¹³ For RTP decisions, the clinician is generally concerned with managing risk of re-injury, but may at times be concerned with risk of any injury, death or illness.

The foundation of the framework is that an injury or re-injury will occur whenever the stress applied to a tissue is greater than the stress the tissue can absorb (tissue health).

Step 1—Tissue health

The first step in the StARRT framework is to assess the stress the tissue can absorb before becoming damaged. This is a function of the health of the tissue. For the same level of activity, the risk of re-injury increases with increasing damage to the tissue that is generally evaluated through the presence of symptoms (e.g. pain), signs

StARRT Framework

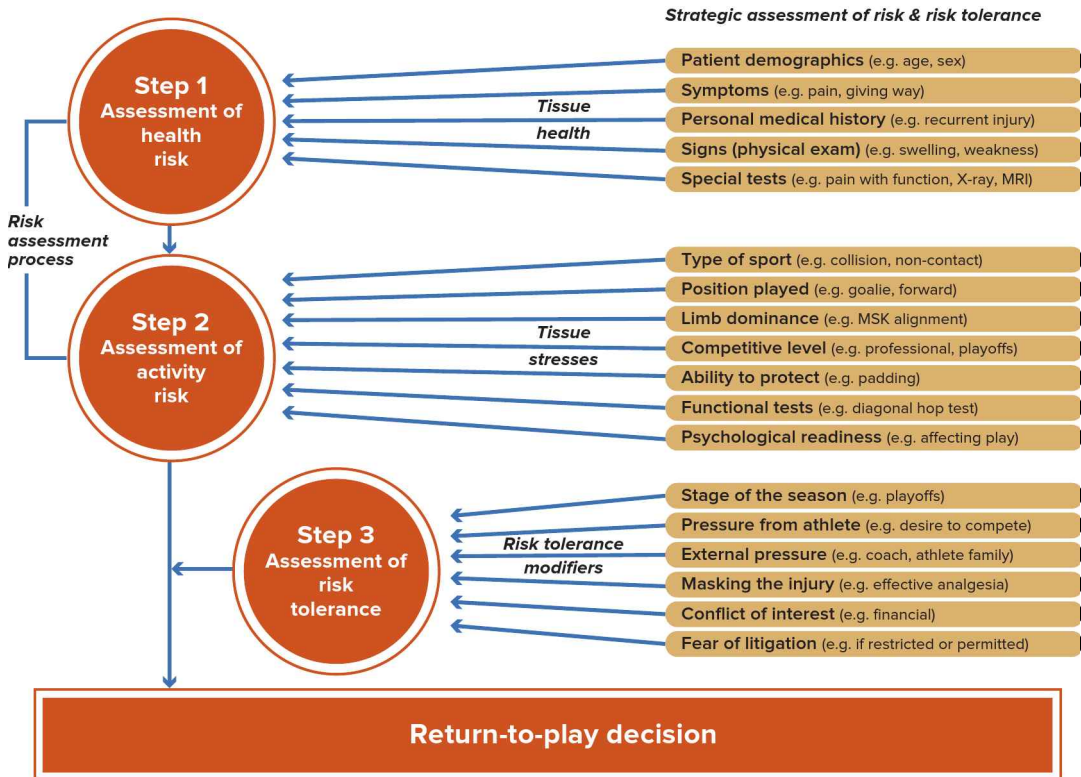


Figure 19.1 The Strategic Assessment of Risk and Risk Tolerance (StARRT) framework for return-to-play (RTP) decisions. This framework illustrates that athletes should be allowed to return to play when the risk assessment (Steps 1 and 2) is below the acceptable risk tolerance threshold (Step 3) and not allowed to return to play if the risk assessment is above the threshold. The StARRT framework groups factors according to their causal relationships with the two components of risk assessment (tissue health, stresses applied to tissue) and risk tolerance, as opposed to the 3-Step framework that groups factors according to the sociological source (medical culture, sport culture, decision maker) of the information. In some cases, an apparently single factor can have more than one causal connection and would be repeated. For example, playoffs will increase the competitive level of play and therefore increase tissue stresses and increase risk. However, playoffs are also expected to affect an athlete's desire to compete (i.e. mood, risk of depression if not allowed to RTP) and could affect financial benefit as well. These causal effects would lead to increased risk tolerance. In this framework, each outcome is evaluated for RTP, and the overall decision is based on the most restricted activity across all outcomes (see text and Table 19.1)

(e.g. swelling) and diagnostic tests (e.g. muscle strength). As this information comes from health professionals, the factors were originally considered 'medical'.

Step 2—Tissue stresses

The second step of the framework is to assess the stress that will be applied to the tissue. This is directly related to the planned activity, as opposed to the status of the original injury and is therefore considered sport-related. Using the frequency, intensity, time and type (FITT) training principle, one can modify activity through frequency (e.g. 3 days per week), intensity (e.g. running fast or climbing hills), timing (e.g. 20 minutes per session) and type. For the type of activity, one can think in general terms such as running or swimming.

However, for re-injury risk, the most salient detail is the required biomechanics that increase stress (i.e. the causes of injury), and the categories 'type of sport' and 'position played' are meant to capture this information. For example, the biomechanics of freestyle swimming are very different from those of breaststroke (analogous to different positions in other sports). Therefore, trying to assess the risk of re-injury in 'swimming' is likely not the best clinical approach. Rather, clinicians should focus on the activities that cause the underlying stresses on the musculoskeletal system to increase. If these can be modified to reduce stress to the injured area (e.g. switch from crawl to breaststroke for some types of shoulder injuries or from whip kick to flutter kick for some types of knee injuries), then the injury risk is decreased.

Psychological factors are listed in Step 2 because, as noted above, they affect risk of re-injury via the way the athlete plays the game and therefore the potential stresses applied to the body. Similarly, functional capacity such as endurance, range of motion and proprioception also describe factors that increase or decrease stresses applied to the injured area and are included here. For example, tight hamstrings mean that forward flexion must come more from the back, which increases lumbar and thoracic stress. Limitations in scapular rotation lead to a decrease in the subacromial space during abduction (increasing the risk of impingement syndrome) as more motion must come from the glenohumeral joint.

Step 3—Risk tolerance modifiers

Step 3 is designed to account for the reasons why two clinicians (or two athletes) presented with the same risk assessment, may come to different RTP decisions. Because the risk is the same, the different RTP decisions reflect differences in risk tolerance. If the competition is the Olympics, a clinician's risk tolerance for the worsening of

a sprained ankle may be greater than if the athlete is a 14-year-old male playing high school basketball. A factor is included as a risk tolerance modifier if there is any context where changing the factor would change one's threshold of an acceptable risk (e.g. for the Olympian) even if it does not affect the decision under other contexts (e.g. high school basketball player).

Return-to-play decision making—beyond risk for injury

The StARRT framework is about arriving at a decision based on whether the assessment of risk exceeds one's risk tolerance. Most clinicians likely think that the framework is about risk for re-injury. Although we are certainly concerned about the risk of re-injury, our focus should be on the overall health or wellbeing of the athlete.¹⁴ How do we define wellbeing or health? Should this be restricted to pathology of muscle, tendon or bone (as for re-injury)? Should we also consider mental health or socioeconomic health?

In a series of studies, we found that sports and exercise medicine clinicians from various fields and regions considered many factors unrelated to risk as important factors in their decisions.⁸ These included several psychological-related factors (desire to compete, psychological impact of not returning to play, loss of competitive standing) as well as potential financial loss, timing of the season, competitive level and fear of litigation. Most interesting was that the pattern of responses in one study that included both clinicians and non-clinicians found very little difference between these two groups.¹⁴

In summary, the StARRT framework organises the information available into risk assessment (Steps 1 and 2) for a particular outcome and information related to other factors that affect one's risk tolerance for that outcome (Step 3) because they may affect other aspects of the athlete's overall wellbeing. In the next section, we discuss how the framework can be applied in some challenging situations.

APPLYING THE StARRT FRAMEWORK

The application of the framework to most clinical situations is straightforward. Consider a case that we used in one of our validation studies where a collegiate American football linebacker (lots of contact at high speed) injured his acromioclavicular joint (Fig. 19.2).¹⁰

As the severity of signs and symptoms increases (from Example 1 through Example 6 in Fig. 19.2), most would agree that the stress the tissue could absorb prior to re-injury would decrease. In our study, as the health of the tissue decreased in the vignettes, clinicians became

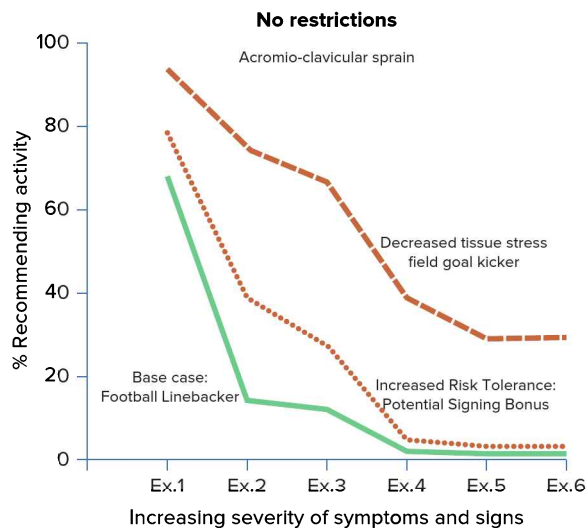


Figure 19.2 The proportion of respondents who would allow activity without restriction is plotted for the different acromioclavicular injury severity levels (increasing severity from Example 1 to 6) described in clinical vignettes.¹⁰ The solid line presents the results for the base case when the athlete is an American football linebacker. The dashed line presents the results when we decreased the likelihood of excessive stress by considering the player to be a field goal kicker instead of a linebacker (tissue stresses). The dotted line presents the results when the linebacker (base case) was being evaluated for a multi-million dollar bonus (risk tolerance modifiers)⁹ REPRODUCED WITH PERMISSION FROM BRITISH JOURNAL OF SPORTS MEDICINE⁹

less likely to allow the athlete to full RTP. We then changed the scenario so that the athlete was a field goal kicker (very low risk of contact) instead of a linebacker. By decreasing the risk of applying excessive stress to the tissue (i.e. contact), the risk of re-injury decreases and this explains why more clinicians allowed full RTP than when the athlete had been a linebacker. Finally, returning to the context of a linebacker, we changed the vignette and said the athlete was being evaluated for a million dollar signing bonus. Given the potential benefit to his economic condition and overall wellbeing, the risk tolerance of many clinicians changed and these clinicians now allowed the athlete to RTP even though the risk of re-injury remained unchanged.

The StARRT framework may appear to discuss RTP decisions as if they are all or none. In clinical practice, RTP may refer to 'full return without restrictions', 'partial return', 'allowed to practice' and so forth. In fact, StARRT is a general framework for any decision-making process and will work equally well when one is trying to decide about return to training, return to practice or any level of activity.

Assessing across outcomes and probabilities

Steps 1 and 2 of StARRT evaluate only a single risk at a time. Yet, RTP decisions usually need to account for many outcomes (e.g. both in the short and long term) simultaneously. In the short term, clinicians and athletes might focus on risk of re-injury or decreased performance (leading to decreased competitive standing). With respect to long-term outcomes, one might focus on risk of osteoarthritis from joint injuries or long-term cognitive deficits following concussions. Each of these represents a different outcome that could be assessed through StARRT. An example of the process for a hypothetical case is shown in Table 19.1. For simplicity, Table 19.1 shows only the overall risk tolerance and omits the particular factors that affect it.

In Table 19.1, Clinician A is the decision maker and evaluates the risk for each outcome. In this hypothetical example, Clinician A considered the risks acceptable and allowed RTP for 6/7 outcomes (column 3), but unacceptable for short-term disability. The *overall RTP decision* must represent the most restricted decision in the table and the athlete is not allowed to RTP. The process is in fact, the same as that which occurs when there is more than one injury (e.g. assessing overall restrictions for an athlete with a knee injury and an ankle injury).

An ankle injury might result in an athlete only partially participating, but a concurrent knee injury might cause the clinician to restrict the athlete's activity to only weight training. The decision-making process for considering five outcomes on one injury in an athlete is the same as that for two outcomes on two injuries in an athlete: evaluate

Table 19.1 Applying the StARRT framework for RTP across different short-term and long-term outcomes. The final decision is the one that imposes the most limiting restrictions of all the outcomes (no RTP in this example)⁹

Outcome	Risk assessment	Risk tolerance	Decision
<i>Short term</i>			
Re-injury	<10%	Acceptable	RTP
Pain	50–70%	Acceptable	RTP
Work disability	20–30%	Not acceptable	No RTP
Performance	20–40%	Acceptable	RTP
<i>Long term</i>			
Osteoarthritis	2%	Acceptable	RTP
Disability	2%	Acceptable	RTP
Cost prohibitive	1%	Acceptable	RTP

across all outcomes and the highest level of restriction becomes the overall restriction.

Risk tolerance modifiers reflect contexts where risk tolerance is affected. These modifiers themselves have particular values (e.g. yes/no or a numerical value if a continuous scale) but are not necessarily risks themselves. For example, timing of the season is pre-season, regular season, finals/playoffs, off-season. In the finals/playoffs, athletes are often playing at higher intensities and more aggressively, leading to increased risk of injury. This is captured in Step 2 under 'competitive level'. In addition, finals/playoffs also have effects on financial compensation, ego and prestige, which are likely to affect risk tolerance in many contexts. Therefore, there may be contexts where timing of the season has minimal effect on risk assessment (Table 19.1, column 2), but large effects on risk tolerance (Table 19.1, column 3) or vice versa.

Baseline risk tolerance simply refers to one's risk tolerance under some arbitrary set of values for the risk tolerance modifying factors. In the case of a knee injury, we might consider a 15% increased risk of re-injury (compared to the uninjured knee) as acceptable. If we believe an athlete's mental health state might be affected by not playing, we might raise the acceptable risk for a knee injury from 15% to 25%. Risk tolerance modifiers are defined by this interaction. Contrast this with the risk of osteoarthritis. If the risk of osteoarthritis is high (or low), we might prevent (or allow) RTP because of the risk of knee osteoarthritis, but it does not change our risk tolerance for the risk of knee re-injury *per se*.

Additional perspectives

The StARRT framework is a simplified version of a more complex process that can lead to decision-tree analysis. When evaluating any action, there must be a comparison with a different action, which may or may not be 'no action'. Table 19.2 illustrates the process for comparing action A (Allow RTP) with action B (Do not allow RTP).

In making a decision, we are hoping for a particular outcome.

Let us consider re-injury as our outcome of interest. If we allow RTP, the athlete may get injured or not get injured. The mental health of the athlete will likely be healthy because they are doing what they want to do. If we do not allow RTP, they cannot get injured. In this context, the mental health of the athlete may be unhealthy (if we are correct that preventing RTP is damaging to the psyche) or may be healthy (if we are not correct that preventing RTP is damaging to the psyche).

In a full decision-tree analysis, one attaches probabilities to the results defined in each row of Table 19.2 (i.e. combination of physical and mental health). Preferably, the probabilities would be based on actual risk scores developed from surveillance or intervention studies (similar to Framingham risk score for cardiovascular disease¹⁵). One could also include the severity of the outcome in terms of disability or decreased capacity. If we assume equal probabilities for each row (25%), one still has to make a value judgment whether being injured with a healthy mental state is better or worse than being uninjured with an unhealthy mental state.

This is a value judgment and there are no algorithms or statistical analyses that can provide the best answer for an individual athlete. Statistically, one multiplies the probability by the value, but this calculation is based on several assumptions that may or may not be true. Regardless, the optimal choice will obviously depend on the disability and long-term consequences of the injury, the disability and long-term consequences associated with the poor mental health and the personal values of whoever is the decision maker.

In reality, the probabilities will not be equal. But the question remains the same and only the proportions change. Is a 20% risk of injury with 10% risk of poor mental health a better choice than 0% chance of being injured with an 80% risk of poor mental health?

Table 19.2 Hypothetical example for an RTP decision-tree analysis approach to RTP. Each action leads to a particular physical state and cognitive state. The probability for the combined physical and cognitive state is estimated, and the 'value' associated with the combined state is given a number between 0 (no value) and 100 (maximum value). The value associated with the state is multiplied by its probability of occurring to provide a total score. The total scores for each row associated with a particular action are summed and the decision is the action that provides the highest decision value

Action	Physical state	Mental state	Probability	Value	Total score (prob*value)	Decision value
Allow RTP	Not injured	Healthy	25%	100	25	37.5
Allow RTP	Injured	Healthy	25%	50	12.5	
No RTP	Not injured	Healthy	25%	60	15	22.5
No RTP	Not injured	Unhealthy	25%	30	7.5	

WHO SHOULD BE THE DECISION MAKER?

Within society at large, most people have the authority to make decisions about their own risks for health unless otherwise restricted by law (e.g. helmets while driving motorcycles in many countries). In the context of returning to play or work after an injury, the decision-making power is often shifted from the individual athlete to the healthcare clinician. For many people, this loss of autonomy can be challenging.

In this section, we discuss different approaches to deciding who should have the authority to make the decision and when and why. A fairly complete list of possible options are: the athlete (or designated representative such as parent or power of attorney), any one of the clinicians (e.g. medical doctor, physiotherapist, athletic trainer, chiropractor, nurse, nutritionist), coach, management (e.g. sport association, team employer), athlete's family, agent or some multidisciplinary team made up of a combination of the above stakeholders. Some of these may be clearly inappropriate in some contexts, but the list is provided to highlight the different factors that need to be considered.

Although power relationships within society are commonly studied in medical sociology,¹⁶ this has not occurred within the context of RTP decisions. The interactions between the athlete, family, friends, agents, coaches, clinicians, and institutional or corporate managers are often complex and confusing. Although clinicians and athletes may 'negotiate' approaches to treatment,¹⁷ the clinician usually has ultimate decision-making power. When athletes avoid interaction with clinicians,¹⁷ they retain power over the decision-making process by default except where the clinician's approval is expressly required.

The first two steps of the StARRT framework require the capacity to evaluate risk. To our knowledge, there is no evidence examining if clinicians are actually better than non-clinicians at evaluating risk of sport injury or whether certain types of clinicians (e.g. medical doctors) are better at evaluating risk compared to other types of clinicians (e.g. physiotherapists).

In one study,¹⁴ we asked members of different Canadian sports medicine professional groups as well as athletes, coaches and sport associations to rank which professional group members might be best to answer such questions. First, most participants felt that medical doctors were the most capable of assessing the health of the tissue. However, when asked who was best to assess the risk of re-injury, short-term and long-term consequences, the members of each of the main clinical professions (medical doctors, physiotherapists, athletic therapists, chiropractors) generally felt that their own professional members had the greatest capacity to evaluate these risks

(e.g. physiotherapists believed physiotherapists were best, chiropractors believed chiropractors were best).

On the other hand, official sport association members generally believed medical doctors were best, athletes believed physiotherapists were often equal to or better than medical doctors, and coaches believed physiotherapists and medical doctors were generally equal but medical doctors were better at assessing long-term consequences. When the same respondents were asked which group had the greatest capacity to assess the importance of risk tolerance modifiers, the athletes and coaches were generally considered the best.

Although these results do not directly speak to who should have the authority to make the RTP decision, they do suggest that an informed decision requires information from several sources. Most importantly, the decision maker needs to act in the best interests of the athlete, which means they have to understand the best interests of the athlete. A coach might have the athlete's best interests at heart, but may not have the experience of watching people suffer with severe osteoarthritis and cannot evaluate the magnitude of disability. A medical doctor may have only limited contact with an athlete and not understand their wishes, dreams or psychological stability.

In addition, as we have seen, there may be differences in opinion about how to weigh all the benefits and risks, but factors that benefit or harm others (i.e. not the athlete) should generally not be included in the decision-making process. Such factors can occur when there are power imbalances. For example, if a clinician is worried about losing his or her status as team doctor or physiotherapist as a result of restricting a player from playing, there may be an unconscious or conscious bias to allow the athlete to RTP even though the risk assessment suggests a risk higher than the decision maker's risk tolerance level in the absence of this fear. Similarly, a coach's performance is based on winning, which might (inappropriately) affect the risk tolerance level even though it is unrelated to the athlete's wellbeing. Table 19.3 summarises some major strengths and weaknesses of each of the stakeholders. The rest of this section discusses some particular points in more detail.

Clinicians

In the workers' compensation injury context, a medical doctor's opinion is generally required to return to work following an injury. This is to protect the worker from unreasonable pressure by the employer. Because the medical doctor is independent, the assumption is that the medical doctor's opinion will be free of bias. This is also the current framework model for most of sports medicine.

However, the external pressures in sport are different from the traditional work environment. In sport, the medical

Table 19.3 Strengths and weaknesses when the decision maker is a particular stakeholder

Stakeholder	Strengths	Weaknesses
<i>Clinicians</i>		
Medical doctor	Knowledge about injury risk and consequences of injury. More often not an employee of management so lower probability of coercion	Less personal contact with individual athletes and may not understand them as well
Team therapist*	Knowledge about injury risk and consequences of injury. Generally know athletes better than medical doctors	Usually an employee of management and may have less perceived power than medical doctor. This might mean more likely subject to coercion
Other (e.g. nutritionist, psychologist)	Similar to medical doctors and usually knowledgeable about injury/illness risk and consequences of injury/illness within their field of expertise	Usually a paid consultant to management. If perceived power is less than that of a medical doctor, then more subject to coercion
<i>Non-clinicians</i>		
Athlete	Knowledge of the athlete's global needs is near complete (but not complete). By definition, provides the best source for information about athlete's needs and values	Only has personal experience about injuries incurred so knowledge base is limited. Difficulty assessing risk. During heightened emotion of competition, may not be able to weigh the risks and benefits properly, even if given information. For injuries affecting cognition (e.g. concussion), informed consent is not possible
Coach	Often aware of athletes' strengths, desires and values. On occasion, will have a more complete view of athlete's benefits such as sport career potential	Limited experience to evaluate risk. During heightened emotion of competition, may be difficult to evaluate risks properly. Potential conflict if athlete may be important to win the game
Family / friends	Often strong knowledge of athlete's global needs, desires and values	Least experience to evaluate risk. On occasion, could be motivated by own personal gains if athlete RTP
Management (sport association / team)	Often aware of career potential and the benefits (or lack thereof) that can occur with RTP	Objective to win may increase risk tolerance even though athlete would not benefit

* The term 'team therapist' is used to refer to a professional who is not a medical doctor and has clinical training to evaluate and treat sport injuries and illnesses. In some cases, the term used to represent a recognised clinical profession in one country represents a completely different (and unrecognised) profession in another country. Team therapists include physiotherapists, athletic therapists, sports therapists, sports rehabilitators, athletic trainers, chiropractors and massage therapists

doctor is usually hired by the team and experiences dual allegiances. Although professional athletes may have access to non-team medical doctors when major conflicts arise, the team medical doctor usually retains the final decision-making power unless legal proceedings ensue. Therefore, unlike the workers' compensation context, medical doctors may indeed be subject to coercion. The same principle is true for all clinicians working for a team. Although financial remuneration will differ with each individual profession, team and region, one should not underestimate the power of social status even when clinical staff are unpaid.

When several health professionals are involved in caring for an athlete, the lines of communication and

authority must be clearly defined.¹ Otherwise, what is already a complex situation for RTP is likely to become even more problematic. According to one consensus document, the essential elements for RTP decisions are the safety of the injured athlete and other athletes, as well as compliance with any rules or regulations.¹ Although some have mentioned that social and economic factors may create pressure for the clinician, the role such factors play in RTP has not been explicitly studied.¹⁸ Because of the absence of clear recommendations, individual treating clinicians have been described as using their own value and belief systems which often include issues other than safety.¹⁷

Athletes

Making RTP decisions requires informed consent. The short-term and long-term consequences of any decision must be evaluated. Therefore, when an athlete's mental status is impaired, he or she would not normally be able to process the information. Although this is obviously the case in certain injuries like concussion, one could also argue that it is the norm in any game or competition context. Simply put, the lack of time available during a competition, accompanied by the emotions that occur during the event, would be expected to affect the ability of almost anyone to process the complex interactions of factors and possible consequences of RTP. However, outside the context of a competition or game, the athlete would normally have the time available to become informed and to evaluate the different consequences with respect to all aspects of health.

What motivates an athlete to want to RTP as soon as possible? In general, the internal factors include (1) that injury may represent a form of body-betrayal and a source of self-resentment,¹⁹ and (2) psychological needs such as the desire to compete or 'love of the game'.²⁰ In addition, the external factors²¹ include sociocultural influences, as well as greater status among peers and fans.^{2,19} Not playing when injured can lead to identity crises, feelings of guilt and shame, or experiences of alienation from the team.²⁰⁻²² All of these factors can lead to low self-esteem and depression.²

In some countries, there is a push towards medical care based on shared decision making for medical conditions.²³ The general concept is that clinicians are sources of information, and help patients through the maze of complex issues and values to achieve greater insight, and therefore participate in the decision-making process. This is in contrast to the traditional model where medical doctors are the authority and patients simply comply. Although some athletes would like to be more involved in RTP decisions themselves,²⁰ others indicate they have learned to trust medical doctors and trainers to make decisions for them,²² and believe it is important for medical personnel to prevent them from returning to sport prematurely.²¹ Conversely, some athletes have reported being pressured into returning to sport too early by the medical doctor, trainer or coach.²¹

The coach

The primary role of the coach is to maximise athletic performance. Some coaches believe athletes can be caught in a 'risk-pain-injury paradox' and that one of the coach's functions is to push athletes to their limits without taking excessive risks.²⁴ This may include competing while injured. Further, some coaches have suggested that medical personnel do not appreciate a need for an individualised approach or that delaying RTP can harm

the athlete psychologically. Coaches also see their role as managing unrealistic athlete expectations about progress following an injury and return to sport.²¹

Family, friends, agents

Family, friends, agents, and institutional or corporate managers all have interests in RTP decisions and their interactions are often complex and confusing. Although it is often believed these persons will place 'non-health issues' above 'health issues', it is apparent that when 'health' is defined very narrowly, it may not represent the best interests of the athlete.

Management

It is clear that management's objective is to win and there is a strong possibility of conflict of interest. That said, management may have important information and perspectives to offer with respect to career potential.

A multidisciplinary approach

If we assume that the RTP decision should be based on the best interests of the athlete, then each of the stakeholders represents a source of information about a different aspect of the athlete. With different personal and societal values, there is clearly not going to be one 'correct' answer for the vast majority of injuries in the vast majority of athletes. In this framework, the key component is to ensure the decision is being based on some combination of the athletes' needs, desires and interests, and that there is no coercion or misrepresentation of information. Could a multidisciplinary model achieve this?

All stakeholders have a potential conflict of interest. Whether this manifests as a real conflict of interest depends on the individuals involved. Although a team approach does not eliminate this risk, its strength is that each team member can help ensure that others' (and their own) potential conflicts of interest are made transparent and documented. This may temper the enthusiasm of some to inappropriately argue in favour of a decision that is mostly for their own self-benefit, as opposed to the wellbeing of the athlete.

There remain significant challenges with a team approach. First, when decisions have to be made very quickly (e.g. during competition), a team approach is not always feasible. Second, a team approach does not eliminate contradicting values, but only brings them together in a transparent fashion. Decisions about risk assessment and risk tolerance can elicit strong emotions and there is a risk that systematising such decisions within a team approach may create interpersonal conflicts that would otherwise be avoided when there is one clear decision maker. Third, when there are conflicting opinions, a mechanism must be available to resolve the issue and come to a decision:

either the athlete returns to activity or does not return to activity.

Should there be a democratic vote with each member having one vote, or should some members' votes have greater weight than others to account for their different capacities and potential conflict of interests? Such a method seems difficult to operationalise, but would have the advantage of providing a significant voice to aspects of athletes' wellbeing that may otherwise go ignored. Exploring solutions to these challenges could be helpful in transforming the current model into one that maximises the chances for athletes' wellbeing to be the focus of the decision.

SUMMARY

In this chapter, we describe the StARRT framework for RTP decision making. The first two steps focus on risk assessment and the third step focuses on factors that alter the decision maker's risk tolerance. If the risk of the outcome is less than the risk tolerance after all factors are

considered, the decision should be to RTP. Otherwise, the decision should be not to RTP.

The current decision-making process is embedded within a social context reflected principally by the values, beliefs and attitudes of a single decision maker. The steps in the process are not always transparent and can lead to confusion and unnecessary conflict. Although the medical doctor is often the ultimate decision maker, many organised sports do not have team doctors and other clinicians make these decisions. Theoretically, each of the stakeholders has information that might be helpful in understanding the best interests of the athlete, and each of the stakeholders is in a potential conflict of interest where their own interests may benefit from one decision or the other.

Finding a solution is likely to be difficult. To borrow a thought from Winston Churchill, at the current time, our current process seems to be the worst form of decision making, 'except for all those other forms that have been tried from time to time'.²⁵ But that doesn't mean we should stop trying yet.

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e

